Beyond Geometric Unity: Infinite Dimensions from One 4D Metric as a Home for Particles

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Abstract

One four-dimensional metric generates an unlimited tower of geometric structure from one short rule, and that tower is a candidate home for the particle spectrum. Four spacetime coordinates plus the metric's ten components at each point give the fourteen-dimensional metric bundle, the starting point of Eric Weinstein's Geometric Unity, and adding derivatives gives 54, 154 and beyond. Once the geometry is given, each level costs only the construction rule and the derivative order, so a theory built on the tower can grow without adding independent inputs. The tower alone is not a particle catalog. A violin string's notes are fixed by its tension, length and clamped ends, and in the same way an action has to say which disturbances of the geometry propagate and how they interact. The paper counts the coordinates, works the oscillator case, and tests one modification, quadratic gravity, against orbits and light bending: its extra forces fade at large distances, so it does not by itself produce a dark halo. Deriving the particles from an action is the next step.

1. One construction, many possible modes

One violin string plays many notes. The particle zoo might work the same way: a small set of rules producing many distinct excitations (field disturbances that appear as particles), with one construction accounting for both the modes, the distinct patterns of vibration, and their interactions.

A string does not acquire its frequencies because its displacement can be differentiated (have its rate of change computed). Its tension, its length and its clamped ends fix the modes. A geometric theory owes the same account. The metric bundle says which configurations are possible; an action, the quantity from which a theory's equations follow, and its solutions say which disturbances propagate and how they interact.

Geometric harmony refers to that explanatory ambition. It does not identify each derivative of the metric, the field that sets distances and durations in spacetime, with another overtone, nor does the existence of an infinite sequence of derivatives show that infinitely many physical excitations are present.

Eric Weinstein's Geometric Unity (Weinstein, 2021) motivates the fourteen-coordinate starting point. Discrete programs, which build spacetime from separate countable pieces, reach relativistic geometry by other routes (Gorard, 2020). The count describes the metric bundle, a space containing possible metrics over spacetime, not fourteen observed spacetime directions. From there, this paper follows one chain: metric components, their derivatives, physical modes selected by dynamics, and a calculable gravitational response.

2. Why the metric bundle has fourteen dimensions

Fourteen is a count of independent numbers: four to say where you are, ten to give the metric there. Stated precisely, let M be a smooth n-dimensional manifold, a space that looks flat up close and supports calculus. At x in M, a metric is a nondegenerate symmetric bilinear form (a rule assigning a number to each pair of vectors) on the tangent space, the space of vectors based at that point. In coordinates, the symmetric matrix has n diagonal and n(n-1)/2 independent off-diagonal entries. Fixing a nondegenerate signature, the count of time-like and space-like directions, leaves the local dimension unchanged. The bundle's

dimension is then

$$\dim E = n + \frac{n(n+1)}{2}.$$

For n=4: four base coordinates and ten fiber coordinates, fourteen in total. A metric field on M is a section of E, one choice of metric at each point. Picking such a choice does not turn the ten fiber coordinates into independent physical excitations. Constraints, gauge equivalence (treating descriptions that differ only in redundant detail as the same) and the equations of motion fix which degrees of freedom, independent ways the system can vary, are physical.

3. Jet bundles: organizing the derivatives

Counting derivatives as extra coordinates makes the list grow fast. The n-th jet bundle (Saunders, 1989) is a list: the metric at a point together with its derivatives up to order n, each entry treated as one more coordinate. In symbols, the number of coordinates is

$$\dim J^n E = 4 + 10\binom{n+4}{4}.$$

The counts begin 14, 54, 154. At first order, each of the ten metric components has four first derivatives, adding 40 coordinates. At second order, symmetry of mixed partials (the order of taking derivatives does not matter) leaves ten second derivatives per component, adding 100. The list lets an action depend on derivatives the way it depends on coordinates. It does not make those derivatives independently excitable physical fields.

One rule builds every level. Choose the order n, run the same jet construction, and the description costs no more than the rule for the bundle, plus about as many bits as it takes to write n in binary and a few more to mark where it ends, plus a fixed amount. Going from 54 coordinates to 154 adds only the bit or two

that says which order. A coordinate list that grows need not make the generating rule grow with it.

As derivative order increases, the counts have no finite upper bound. Carry compatible derivative data at every finite order and the result is the infinite jet tower. Its base remains four-dimensional spacetime; the additional coordinates describe field values and derivatives, not new spacetime directions. A finite rule specifies the hierarchy without listing infinitely many coordinates separately (De Vecchi, 2017).

An infinite derivative list at a point does not generally determine a smooth function (one with derivatives of every order) away from that point. The function exp(-1/x squared) for nonzero x, zero at x=0, is smooth with all derivatives vanishing at zero, yet nonzero elsewhere. For an analytic function, a convergent Taylor expansion, a sum built from the function's derivatives at one point, determines the function in a neighborhood. Rebuilding a function from its Taylor series, or coding a geometric object, is not the same thing as a physical holographic duality (a theory in a region equivalent to one on its boundary), which has to relate observables and dynamics.

4. The description cost of generated geometry

Running one fixed algorithm costs a constant, however large its output (Solomonoff, 1964). Stated precisely, for a fixed effective procedure F (a mechanical step-by-step method) that builds the metric bundle from a finite description D, $K(F(D)|\, D) \leq c_F$ and $K(D,F(D)) \leq K(D)+c'_F$, where the constants depend on the procedure and reference machine (the standard computer used to measure description length), not on the particular input D. A bounded reusable cost need not be zero, and a section, an action or a boundary condition (a requirement fixed at the edges of a region) you choose is not automatically an

output of F.

Stopping at a finite order adds one charge, the order itself. Written out, for a finite jet construction F_n a fixed algorithm reads both D and an encoded order n, giving K(F_n(D) | D,n) bounded by a constant. Increasing the order adds the cost of specifying n.

5. The same geometry can support different laws

Two theories can share every coordinate and still disagree about the vacuum, spacetime empty of matter. Written out on a four-dimensional Lorentzian base (a spacetime whose geometry treats time differently from space), the two actions are

$$S_0[g] = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\,R,$$

$$S_\Lambda[g] = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\,(R-2\Lambda).$$

The corresponding vacuum equations are G_ab=0 and G_ab+Lambda g_ab=0. Both theories use the same metric configuration bundle. A flat metric, one for uncurved spacetime, solves the first; a nonzero Lambda, the cosmological constant giving empty space an energy, excludes that flat vacuum in the second. Fourteen coordinates therefore do not select even this elementary difference in dynamics. Observation decides which action describes a physical system.

6. Which disturbances become physical modes?

Differentiating one oscillatory function changes its amplitude and phase (size and timing within the cycle) in a prescribed way; it does not produce a new independent oscillator at a new frequency for every derivative. Physical modes come from a kinetic operator (the part of the action that governs how disturbances

travel) and its constraints about a chosen background, a fixed solution around which disturbances are studied. A higher-derivative action may introduce extra poles (frequencies at which the response becomes infinite, each marking a mode) or pathological modes; it need not add one healthy mode per derivative order.

One oscillator, infinitely many derivatives

A list of derivatives is not a list of modes, and one cosine shows why. In symbols, let q(t) = A cos(omega t). Its n-th derivative is A omega^n cos(omega t + n pi/2). Every derivative has the same frequency. Treating q, dot q, double-dot q and so on as a list creates infinitely many mathematical entries without creating infinitely many freely excitable oscillators.

An extra frequency has to be put in by hand, through a term in the action. Written out, the illustrative mechanical action is

$$S[q] = \frac{1}{2}\int dt\left[\dot{q}^2 - \omega^2 q^2 - \eta\ddot{q}^2\right], \qquad \eta > 0.$$

The Euler-Lagrange equation, the equation of motion an action yields, is eta q^(4) + double-dot q + omega^2 q = 0. Substituting q proportional to exp(i nu t) gives two squared frequencies:

$$\nu_{\pm}^2 = \frac{1 \pm \sqrt{1 - 4\eta\omega^2}}{2\eta}.$$

The extra frequency comes from eta and the action, not from q having derivatives. As eta tends to zero, the lower frequency tends to omega and the higher diverges (grows without limit). The inverse of that quadratic operator has opposite residues (the strengths of its pole contributions) at its two frequency poles, so real frequencies do not by themselves establish a positive-energy physical spectrum.

Infinite smoothness is compatible with a second-order equation (one involving at most second derivatives) having only a small number of physical modes. Conversely, one higher-derivative term can introduce an unwanted mode. The instrument is the dynamics, not the derivative list.

7. The action fixes the particle content

The fermion question

A metric-jet hierarchy does not automatically supply spinors, the kind of field an electron is. Covariant quantities formed from the metric, curvature and their tensor derivatives transform as tensors, generalizations of vectors with fixed transformation rules. A 2pi spatial rotation, one full turn, acts trivially on them, whereas it acts by a minus sign on a spinor field. Relabeling a derivative coordinate cannot change that transformation law.

The obstruction blocks one route: calling a tensor field a fundamental spinor. Fermions, particles like electrons that cannot share a quantum state, can emerge from bosonic systems, made of particles that can share a quantum state (Levin and Wen, 2003), which is additional physics a proposal would have to exhibit.

The geometric proposal therefore needs either spinor fields whose representations and couplings (how they transform under symmetries and interact) it explains, or dynamics that produces fermionic excitations. Deriving the Standard Model, the established theory of known particles and forces except gravity, means recovering chiral (handedness-dependent) matter representations, gauge interactions and anomaly cancellation, the cancelling out of quantum effects that would break symmetries. If those ingredients are supplied independently, they belong in the complete description cost.

A geometric operator expansion

Writing the general action shows how many free coefficients the coordinate count leaves open. In symbols, a possible four-dimensional effective action, an approximate action valid at low energies, begins with

$$S = \int d^4x \sqrt{-g} \left[\frac{M_P^2}{2} R - \Lambda_{\text{vac}} + \alpha R^2 + \beta C^2 + \frac{\gamma}{M^2} R \Box R + \cdots \right] + S_m .$$

M is a suppression scale, the energy at which the extra terms become important; alpha, beta and gamma are coefficients; S_m determines matter and its coupling. The jet coordinate count neither selects these coefficients nor sets Lambda_vac. Operator mass dimension (the power of mass units a term carries) and jet order count different things; products of fields and derivatives yield many operators at the same order, subject to identities and redundancies.

Geometric harmony has a definite task: derive a restricted operator pattern with a controlled low-energy regime, rather than call every possible coefficient an amplitude available to fit afterward. A successful result could predict ratios or forbid terms that a generic expansion allows.

Stability and prescription

Removing an unwanted mode is a physical choice with its own costs. Damiano Anselmi's fakeon prescription is one approach to the unitarity problem, the question of whether total probability is preserved: it excludes selected modes from the asymptotic physical states, the states that can arrive at a detector, at the cost of violating microcausality, cause preceding effect even at the smallest scales (Anselmi, 2019). A mode that does not arrive as a particle is not thereby a dark-matter fluid that clusters on its own. Which prescription is chosen, what classical limit (behavior once quantum effects are negligible) it has and what scales it carries are physical content, and each must be assessed. This paper

does not identify every higher-derivative mode with a fakeon or every fakeon with dark matter.

8. Dark matter: a worked gravitational response

Quadratic gravity's extra forces, produced by added terms that square the curvature, fade with distance (Stelle, 1977; Alvarez-Gaume et al., 2015), so a point source leaves orbital speeds falling off. The benchmark carries positive inverse lengths m0 and m2 for the scalar and spin-2 contributions, so that 1/m0 and 1/m2 are the ranges over which those two contributions act. Written out, the standard static weak-field point-source potential (Burzilla et al., 2021) is

$$\Phi(r) = -\frac{GM}{r}\left[1 + \frac{1}{3}e^{-m_0 r} - \frac{4}{3}e^{-m_2 r}\right].$$

For a circular orbit, v_c squared equals r times the radial derivative:

$$v_c^2(r) = \frac{GM}{r}\left[1 + \frac{1}{3}(1+m_0 r)e^{-m_0 r} - \frac{4}{3}(1+m_2 r)e^{-m_2 r}\right].$$

Both exponential terms vanish as r grows, so v_c approaches zero. This potential does not generate an asymptotically flat rotation curve, orbital speeds leveling off at large distances, for a finite point-source mass. A useful fit over a finite radial interval is not excluded.

Finite-radius bound

For a point source the effect is capped: the two extra terms cannot raise the circular-speed factor above 4/3. Stated precisely, write f(x)=(1+x)exp(-x). For x>=0, f(0)=1 and f is nonincreasing, so 0<=f(x)<=1. The point-source circular-speed factor is

F(r)=r v_c^2/(GM)=1+(1/3)f(m_0r)-(4/3)f(m_2r)<=4/3.

With fixed source mass and G, the model cannot supply a dynamical mass enhancement, a boost in the mass inferred from orbits, larger than one third at any radius. This bound cannot be transplanted to the interior of an extended disk, where the Yukawa force, a force fading exponentially beyond a set range, must be integrated over the source.

Extended sources and lensing

Light bending and orbital motion respond to different combinations of the two potentials (Bartelmann and Schneider, 2001). Written out in the weak-field convention ds^2=-(1+2 Phi)dt^2+(1-2 Psi)dx^2, the extended-source potentials in this benchmark are

$$\Phi(\mathbf{x}) = -G\int d^3x' \frac{\rho(\mathbf{x}')}{s}\left[1 + \frac{1}{3}e^{-m_0 s} - \frac{4}{3}e^{-m_2 s}\right],$$

$$\Psi(\mathbf{x}) = -G\int d^3x' \frac{\rho(\mathbf{x}')}{s}\left[1 - \frac{1}{3}e^{-m_0 s} - \frac{2}{3}e^{-m_2 s}\right].$$

The lensing sum, the combination that governs light bending, is

$$\Phi + \Psi = -2G\int d^3x' \frac{\rho(\mathbf{x}')}{s}[1 - e^{-m_2 s}].$$

The scalar Yukawa cancels in this sum although it remains in Phi. A lensing fit cannot be inferred from a rotation-curve fit. Outside a source with bounded support, one confined to a finite region, the finite-range terms decay; the ordinary far-field mass term survives.

Effective density

The modification moves dynamical mass around inside a source without adding to the total. Apply the distributional identities for the Laplacian (a sum of second

derivatives over all directions) of 1/s and of exp(-ms)/s, and note that 1+1/3-4/3=0, so the contact terms, contributions concentrated at the source point itself, cancel. In symbols, the effective dynamical density is

$$\rho_{\mathrm{eff}}(\mathbf{x}) = \frac{1}{4\pi}\int d^3x' \frac{\rho(\mathbf{x}')}{s}\left[\frac{4}{3}m_2^2 e^{-m_2 s} - \frac{1}{3}m_0^2 e^{-m_0 s}\right].$$

This is a convolution (a smearing of the source by a distance-dependent weighting) whose sign and profile depend on both inverse ranges and the source. The total integral of rho_eff for a localized source equals M, and the effective excess rho_eff minus rho integrates to zero. This benchmark redistributes dynamical density but does not create extra total far-field mass.

Observational test

One set of parameters has to fit both orbital speeds and light deflection. Start with a galaxy's baryonic density, the density of ordinary matter, inferred from stars and gas. Evaluate Phi in the disk plane and differentiate it to obtain v_c^2(R) at the measured radii. State whether m_0 and m_2 must be shared across galaxies. For a system with lensing data, use the same parameters to evaluate the line-of-sight derivative of Phi+Psi.

Uncertain stellar mass-to-light ratios, distances, inclinations and gas distributions affect both comparisons and belong in the likelihood, the probability of the data given the model. The SPARC catalog, Spitzer Photometry and Accurate Rotation Curves for 175 disk galaxies, supplies rotation-curve data (Lelli, McGaugh and Schombert, 2016). No SPARC fit is reported here; the calculation specifies a test the proposed mechanism must pass.

9. Comparing geometric and particle accounts

A fair comparison charges each account for everything it needs. Fix the scope: observables, precision, preparation conditions and boundary data. Count the geometry, fields, equations of motion, boundary conditions and observation rule together. Their joint description length can be shorter than separate descriptions because one ingredient may determine another.

If one construction derives relations among couplings that a rival must specify separately, those relations reduce the joint cost. Count the derivation once and do not charge its consequences again. Conversely, an unspecified action cannot receive a zero cost simply because someone has defined a rich configuration space.

A particular particle model can relate the mass, interaction strength and production history of a WIMP, a weakly interacting massive particle, or an axion's mass and couplings through its underlying construction. Assigning each a generic bit cost while treating a new gravitational kernel, the rule weighting a source's pull by distance, as free does not compare complete explanations. The proposed metric theory must pay for m_0, m_2, its action and its prescription at whatever precision the same observations require.

Description length converts into odds only when the saving has been established. Under the description prior, in which each bit is a 50/50 choice that no rule makes, and with equal observational likelihoods, a demonstrated delta-bit saving gives a factor 2^delta in prior and posterior odds, the odds before and after the evidence. Establishing the saving is the outstanding comparison. The fixed jet-construction algorithm can be shared across geometric candidates, and it should not be mistaken for the physical content of any one of them.

10. From the construction to a physical theory

Specify an action and background, derive the independent propagating modes, and check whether disturbances grow without bound. Then determine their coupling to matter and which parameters the construction fixes.

The first observational target is a joint account of orbital motion and lensing using the same coefficients. A galaxy fit would not by itself establish agreement with cluster mergers (collisions of galaxy clusters), cosmology (Planck Collaboration, 2020) or gravitational waves. Those are subsequent tests, not results claimed here. The present contribution is the geometric hierarchy and a worked gravitational response exposing a concrete limitation of the finite-range example.

11. Conclusion

The fourteen-dimensional metric bundle and its finite jets come from specified geometric constructions, and their growing coordinate lists can share one short generating rule. An action, its constraints and a background then fix which modes are physical.

The quadratic-gravity calculation supplies a definite test. With the stated positive inverse ranges, its additional forces decay at large distance, its effective total mass equals the source mass, and its lensing potential differs from its orbital potential. A finite-range fit may still be investigated, but the construction alone is not a derived dark halo.

Geometric unification becomes explanatory when one action supplies relations among particles, interactions and observations that otherwise require independent inputs. This paper supplies the coordinate counting, the mode calculation and the gravitational response. Deriving particles from an action is the next step.

Appendix A. Why the point-source bound does not extend to every source

A.1 Point-source result restated

The point-source result sets the ceiling the shell calculation will exceed. Write f(x) = (1 + x)e^(-x). For x >= 0, f is nonincreasing with f(0) = 1, so 0 <= f(x) <= 1. The point-source circular-speed factor from Section 8 is

$$F(r) = \frac{r\,v_c^2}{GM} = 1 + \tfrac{1}{3} f(m_0 r) - \tfrac{4}{3} f(m_2 r) \le \tfrac{4}{3}.$$

A.2 Outer shell

An outer shell can exceed the point-source bound because its Newtonian force vanishes inside while its Yukawa force need not. Stated precisely, consider a thin spherical shell of radius a and mass M_s around a central point mass M_c. For 0<r<a, averaging the kernel -G exp(-ms)/s over the shell's angles gives the interior Yukawa potential

$$Y_m(r) = -\frac{GM_s\, e^{-ma}}{mar} \sinh(mr), \tag{1}$$

with radial derivative

$$r\,Y_m'(r) = -\frac{GM_s\, e^{-ma}}{mar} [mr\cosh(mr) - \sinh(mr)] < 0. \tag{2}$$

The circular speed from the shell's contribution is

$$v_{c,\,\mathrm{shell}}^2(r) = \tfrac{1}{3}\, r\, Y'_{m_0}(r) - \tfrac{4}{3}\, r\, Y'_{m_2}(r). \tag{3}$$

Because the Yukawa kernel integrates over the source rather than being evaluated at a point, the interior response of the shell is not bounded by the point-source formula.

A.3 Numerical instance

A single numerical case breaks the point-source bound. Take r = 1, a = 2, m_0 = 5, m_2 = 1, M_c = 1, M_s = 100. Evaluating the central mass plus shell contributions, with v_N the Newtonian circular speed of the same masses, which inside the shell comes from the central mass alone:

$$v_c^2/v_N^2 \approx 3.307, \tag{4}$$

which exceeds the point-source bound of 4/3. The surplus comes from the shell integral, whose sinh kernel amplifies the longer-range spin-2 Yukawa contribution when the shell lies at a favorable ratio of a to 1/m. Here that term supplies +3.319 while the shorter-range scalar term, suppressed by e^(-10), contributes -0.045.

This confirms that the Section 8 bound applies to point sources and cannot be transplanted to extended mass distributions without recalculating the source integral.

A.4 Reproducibility

The supplement A18_Shell_Check.py evaluates equations (1) through (4) with the standard library. It checks the shell average in (1) by integrating the kernel over the shell, the derivative in (2) by finite differences (numerical slopes taken over small steps) and the point-source ceiling of 4/3 on a grid, and it reproduces the ratio in (4), 3.3067, with its spin-2 term of +3.3191 and its scalar term of -0.0449. Its output is A18_Shell_Check_Results.json.

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